

Genistein inhibits differentiation of primary human adipocytes.

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Genistein, a major soy isoflavone, has been reported to exhibit antiadipogenic and proapoptotic potential in vivo and in vitro. It is also a phytoestrogen which has high affinity to estrogen receptor beta. In this study, we determined the effect of genistein on adipogenesis and estrogen receptor (ER) alpha and beta expression during differentiation in primary human preadipocytes. Genistein inhibited lipid accumulation in a dose-dependent manner at concentrations of 6.25 microM and higher, with 50 microM genistein inhibiting lipid accumulation almost completely. Low concentrations of genistein (3.25 microM) increased cell viability and higher concentrations (25 and 50 microM) decreased it by 16.48+/-1.35% ($P<.0001$) and 50.68+/-1.34% ($P<.0001$). Oil Red O staining was used to confirm the effects on lipid accumulation. The inhibition of lipid accumulation was associated with inhibition of glycerol-3-phosphate dehydrogenase activity and down-regulation of expression of adipocyte-specific genes, including peroxisome proliferator-activated receptor gamma, CCAAT/enhancer binding protein alpha, glycerol-3-phosphate dehydrogenase, adipocyte fatty acid binding protein, fatty acid synthase, sterol regulatory element-binding protein 1, perilipin, leptin, lipoprotein lipase and hormone-sensitive lipase. These effects of genistein during the differentiation period were associated with down-regulation of ERalpha and ERbeta expression. This study adds to the elucidation of the molecular pathways involved in the inhibition of adipogenesis by phytoestrogens.